



Fig. 20: TWI write menu

for acknowledging the reception. An acknowledge signal (ACK) is sent back by the receiver, pulling the SDA line low during the ninth SCL cycle. If the receiver leaves the SDA line high, an NACK is signalled. When the receiver has received the last byte, or for some

reason cannot receive any more byte, it should inform the transmitter by sending an NACK after the final byte. The MSB of the data byte is transmitted first.

For details of the TWI module, addressing mode, transmit mode, receive mode, etc, you may go through the datasheet of ATmega32.

The twi.c source code also uses the USART module to provide a user interface from the PC in order to perform various tasks like writing, reading or erasing data on the EEPROM chip. To set up the HyperTerminal in your PC, follow the instructions given in the previous

section.

In the source code, the TWI module is configured to operate at a clock frequency of 100 kHz with no prescaler. The USART module is configured to operate at a baud rate of 57,600 bps with standard frame format of 8 data bits, one stop bit and no parity bit.

If you have loaded the code in the MCU successfully, you will see a welcome screen with "Press any key to continue" message. Screenshots of the HyperTerminal at different points of executions are shown in Figs 18, 19 and 20.

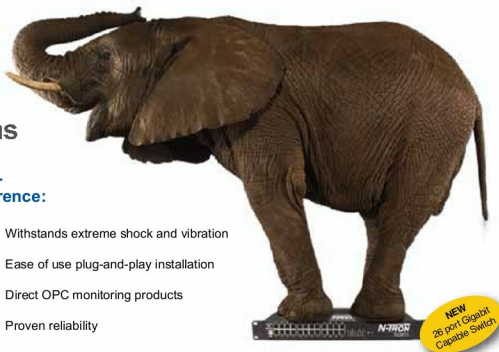
EFY note. All the example codes and reference manual of this article are included in this month's EFY-CD and are also available on the www.efymag.com website. ●

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